Question Paper

Exam Date & Time: 14-Jun-2024 (02:30 PM - 05:30 PM)

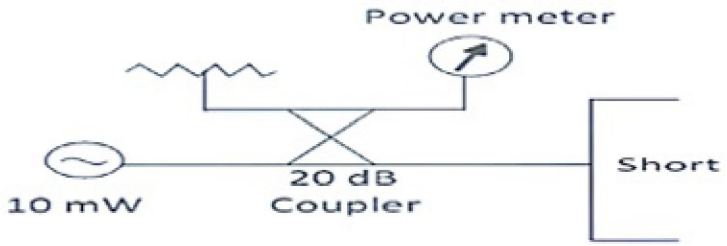
MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING) DEGREE EXAMINATIONS - JUNE 2024
SUBJECT: ECE 2224/ECE_2224 - MICROWAVE ENGINEERING

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) In A Microwave signal for a directional coupler measurement is given in the figure. Here assume that the calculated power from the meter is 0.01 μW. Find the coupler directivity. (4)



1B) 40mW microwave signal is fed to arm 3 of E plane tee by a match terminator. Arms 1 and 2 are terminated respectively in 40Ω and 60Ω loads, and the wave impedance of the guide is 50Ω. Find the power delivered to the loads. (4)

1C) A three-port circulator has an insertion loss of 2dB, Isolation of 30dB, and VSWR is 1.3. Find its scattering matrix. Assume the phases of all coefficients are zero. (2)

2A) Explain the operation of a circulator using a suitable diagram. (4)

2B) With a detailed diagram, demonstrate the construction and working principle of Klystron. (4)

2C) List out the applications of TWT. (2)

3A) Find the directivity of an infinitesimal dipole antenna. (4)

3B) Explain the working of Two-hole directional coupler and write its corresponding S- Matrix. (4)

3C) Calculate the radiation efficiency of a lossless dipole antenna. (2)

4A) Derive the vector wave equation in terms of magnetic vector potential and obtain the solution of it. (4)

4B) For an infinitesimal dipole determine and interpret the vector effective length. At what incidence angle does the open circuit maximum voltage occurs at the output terminals of the dipole if the electric field intensity of the incident wave is 10 mV/m? The length of dipole is 10 cm. (4)

4C) Define (i) Directivity (ii) Radiation resistance of the antenna. (2)

5A) With the necessary diagrams and equations, explain in detail (a) Folded dipole and (b) Helical antenna. (4)

5B) With the necessary diagrams and equations, explain how the microstrip circular patch antenna works. (4)

5C) Sketch the formation and detachment of electric field lines for a short dipole. (2)
(i) t=T/4 (ii) $t = T/2$ (iii) t=3T/4. Where T is time period.

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